MILITARY SPECIFICATION SHEET

*ELECTRON TUBES, TRANSMITTING*

TYPES 6146W AND 6159W 1/
a b

The complete requirements for procuring the electron tubes described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Beam power amplifier, F1 = 60 MHz, F2 = 175 MHz

Outline --- See figure 1

Cathode --- Coated unipotential

Base connections:

| Pin No. | --- | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Cap |
|---|---|---|---|---|---|---|---|---|---|---|
| Element | --- | k, int sd, g3 | h | g2 | k, int sd, g3 | g1 | k, int sd, g3 | h | Base sleeve | a |

ABSOLUTE MAXIMUM RATINGS:

| Parameter: | Test code | Ef | Eb | Ec1 | Ec2 | Ib | Ic1 | Pp | Pg2 | Pi | Ehk | Modulation | Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | | V | Vdc | Vdc | Vdc | mAdc | mAdc | W | W | W | Vdc | | ft |
| Class AB1, AF: | a | 6.3±10% | 600 | --- | 250 | 125 | --- | 20 | 3 | 60 | ±135 | --- | 10,000 |
| Class AB1, AF: | b | 26.5±10% | 600 | --- | 250 | 125 | --- | 20 | 3 | 60 | ±135 | --- | 10,000 |
| Class AB2, AF: | a | 6.3±10% | 600 | --- | 250 | 125 | --- | 20 | 3 | 62.5 | ±135 | --- | 10,000 |
| Class AB2, AF: | b | 26.5±10% | 600 | --- | 250 | 125 | --- | 20 | 3 | 62.5 | ±135 | --- | 10,000 |
| Class C, Telep: | a | 6.3±10% | 480 (see fig 2) | -150 | 250 | 117 | 3.5 | 13.3 | 2 | 45 (see fig 2) | ±135 | Anode | 10,000 |
| Class C, Telep: | b | 26.5±10% | 480 (see fig 2) | -150 | 250 | 117 | 3.5 | 13.3 | 2 | 45 (see fig 2) | ±135 | Anode | 10,000 |
| Class C, Teleg: | a | 6.3±10% | 600 (see fig 2) | -150 | 250 | 140 | 3.5 | 20 | 3 | 67.5 (see fig 2) | ±135 | --- | 10,000 |
| Class C, Teleg: | b | 26.5±10% | 600 (see fig 2) | -150 | 250 | 140 | 3.5 | 20 | 3 | 67.5 (see fig 2) | ±135 | --- | 10,000 |
| TEST CONDITIONS: | a | 6.3 | 300 | --- | 200 | --- | --- | --- | --- | --- | --- | --- | --- |
| TEST CONDITIONS: | b | 26.5 | 300 | --- | 200 | --- | --- | --- | --- | --- | --- | --- | --- |

GENERAL:

Qualification - Required

1/ To identify those tests that are applicable to a given type or to several types; tube types are designated by letters.

Ⓐ denotes changes



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| METHOD | REQUIREMENT OR TEST | TYPE | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| Ⓐ | Qualification inspection | | | | | | | | |
| 1131 | Metallic base sleeve quality | a, b | | --- | --- | --- | --- | --- | --- |
| 1216 | Base material insulating quality | a, b | Zone 11 (min) | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 1 | | | | | | | | |
| 1231 | Emission | a, b | Eb = Ec1 = Ec2 = 20 Vdc (see note 1) | 0.65 | II | Is | 110 | --- | mAdc |
| 1256 | Electrode current (1) (anode) | a, b | Ec1 = -33 Vdc | 0.65 | II | Ib | 46 | 94 | mAdc |
| 1266 | Total grid current | a, b | Ec1/Ib = 67 mAdc; t = 120 (see note 1) | 0.65 | II | Ic1 | --- | -3.0 | μAdc |
| 1256 | Electrode current (screen) | a, b | Ec1 = -33 Vdc | 0.65 | II | Ic2 | 0 | 5.5 | mAdc |
| Ⓐ --- | Short and discontinuity detection | a, b | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | | |
| 1031 | Low-frequency vibration | a, b | Eb = 250 Vdc; Ec1/Ib = 10 mAdc; Rl = 2,000 ohms | --- | --- | Ep | --- | 500 | mVac |
| 1031 | Variable-frequency vibration (1) | a, b | Low-frequency vibration | --- | --- | Ep | --- | 500 | mVac |
| --- | Variable-frequency vibration (2) | a, b | Low-frequency vibration (see note 4) | --- | --- | Ep | --- | 500 | mVac |
| 1301 | Heater current | a | | --- | --- | If | 1.175 | 1.325 | A |
| 1301 | Heater current | b | | --- | --- | If | 0.28 | 0.32 | A |
| 1336 | Heater-cathode leakage | a, b | | --- | --- | Ihk | --- | 100 | μAdc |
| 1256 | Electrode current (2) (anode) | a, b | Ec1 = -70 Vdc | --- | --- | Ib | 0 | 2 | mAdc |
| 1306 | Transconductance | a, b | Eb = 200 Vdc; Ec1/Ib = 100 mAdc | --- | --- | Sm | 5,600 | 8,400 | μmhos |
| 1331 | Direct-interelectrode capacitance | a, b | No shield, base sleeve tied to cathode | --- | --- | Cgp<br>Cin<br>Cout | ---<br>12.0<br>7.3 | 0.24<br>15.0<br>9.5 | pF<br>pF<br>pF |
| 1236 | Power oscillation | a, b | Eb = 600 Vdc; Ec2 = 180 Vdc; Ib = 112 mAdc (max); Rg1 = 0.03 Meg; Ic1 = 2.0 to 2.5 mAdc; F = 15 MHz | --- | --- | Po | 47 | --- | W (useful) |



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| METHOD | REQUIREMENT OR TEST | TYPE | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 2 -Continued | | | | | | | | |
| 1031 | Vibration fatigue | a, b | See note 3 | 6.5 | S2 | --- | --- | --- | --- |
| 1041 | Shock | a, b | 450 G; rated heater voltage applied; no other voltages | --- | --- | --- | --- | --- | --- |
| --- | Post-shock and vibration fatigue test end points | a, b | Low-frequency vibration | --- | --- | Ep | --- | 500 | mVac |
| | | | Power oscillation | --- | --- | Po | 42 | --- | W |
| | | | Heater-cathode leakage | --- | --- | Ihk | --- | 100 | µAdc |
| (A) 1101 | Secureness of base, cap, or insert | a, b | | --- | --- | --- | --- | --- | --- |
| (A) 1105 | Permanence of marking | a, b | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | | |
| --- | Life-test provisions | a, b | Group C; Eb = 400 Vdc; Ib = 260 mAdc; Ic1 = 4.0 to 5.0 mAdc; F = 125 MHz; Rg1 = 0.015 Meg (see note 2) | --- | --- | --- | --- | --- | --- |
| --- | Life-test end points (500 hours) | a, b | Emission | --- | --- | Is | 90 | --- | mAdc |
| | | | Power oscillation | --- | --- | Po | 42 | --- | W |

NOTES:

1. This test to be performed at the conclusion of the holding period.
2. Test in a self-excited push-pull oscillator circuit. Current and grid resistance values specified are for two tubes.
3. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes.
4. Vibrate at 10 G in each of three positions X, Y, and Z through the frequency range from 50 to 120 Hz. The time for gradually covering the range from 50 to 120 Hz shall be 3 to 15 minutes.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army -
Air Force - 11
DSA - ES

User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 19

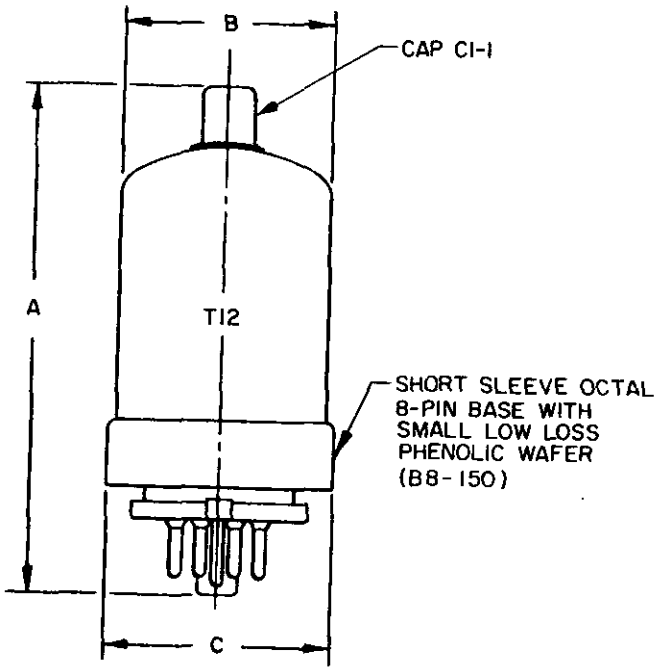
Preparing activity: Navy - EC

Agent: DSA - ES

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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses | |
|---|---|---|
| | Minimum | Maximum |
| Quality conformance inspection, part 2 | | |
| A | 3.562 (90.47) | 3.812 (96.82) |
| B | | 1.562 (39.67) |
| C | | 1.656 (42.06) |

FIGURE 1. Outline drawing of electron tube types 6146W and 6159W.



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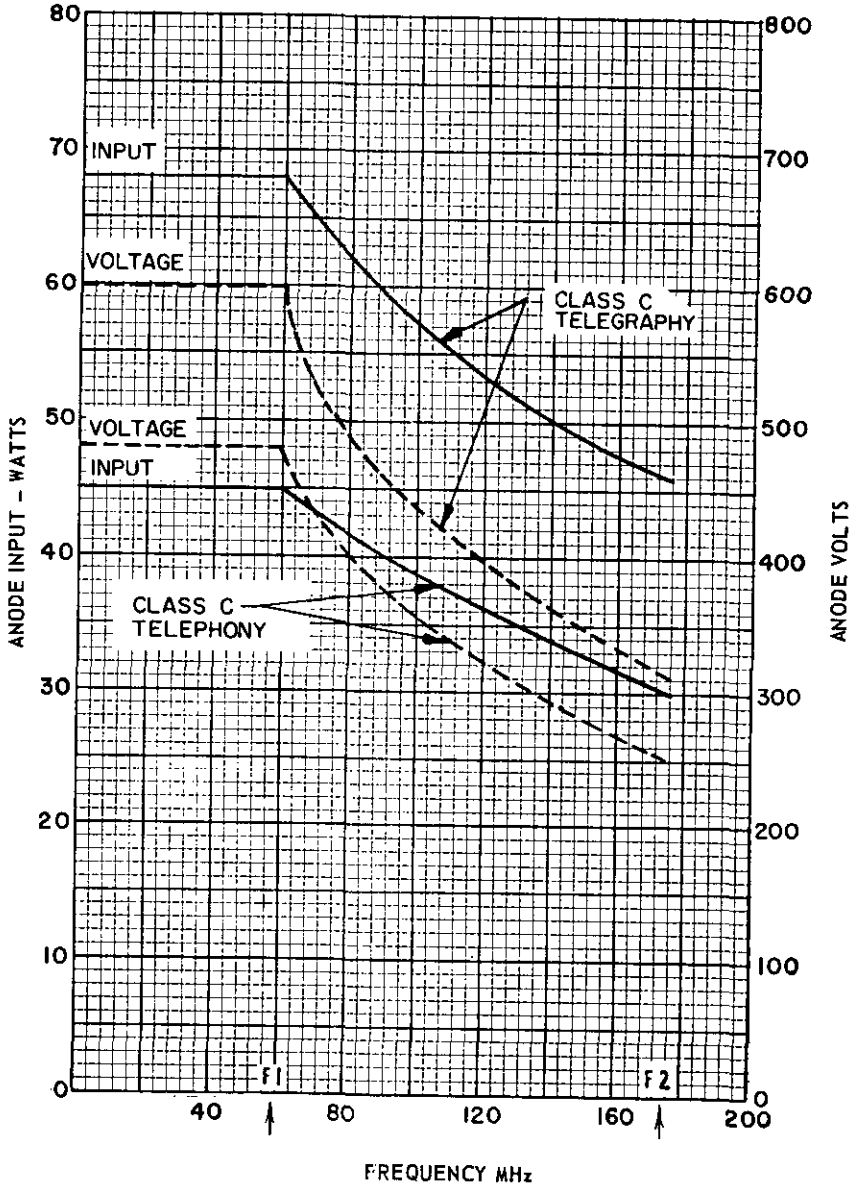



FIGURE 2. Anode voltage and anode input versus frequency for types 6146W and 6159W.



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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